

Program : Diploma in Communication and Computer Networking	
Course Code : 5328	Course Title: Application Software Development Lab
Semester : 5	Credits: 1.5
Course Category: Programme core course	
Periods per week: 3(L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Familiarise Open source Tools: StarUML / UMLGraph / Topcased/Umberollo for modeling
- Prepare a SRS document in line with the IEEE recommended standards.
- Develop medium-sized project in a team

Course Prerequisites:

Topic	Program / Course Name
Software engineering concepts	Software Management System

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Prepare a SRS document in line with the IEEE recommended standards	8	Applying
CO2	Modelling UML Behavioural Diagrams and Capturing different Scenarios	12	Applying
CO3	Modelling using UML structural diagrams	9	Applying
CO4	Develop medium-sized project in a team	8	Applying
	Series Test	4	
	Open Ended Experiments / micro projects	4	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			1
CO2	3			3			1
CO3	3			3			1
CO4	3			3	2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identifying the Requirements and prepare SRS		
M1.01	Analyze a real problem situation/application. Write down the problem statement for a suggested system of relevance.	3	Applying
M1.02	Do requirement analysis and develop Software Requirement Specification Sheet (SRS) for suggested system.	5	Applying
CO2	Modeling UML Behavioural Diagrams and Capturing different Scenarios		
M2.01	Develop use case diagram,by identifying Actors ,Use cases Association between Actors and Use Cases , Use Case Relationships ,Include Relationship , Extend Relationship	4	Applying
M2.02	Statechart and Activity Modeling	4	Applying
M2.03	Modeling behavioral aspects-Sequence and Collaboration diagram	4	Applying
	Series Test– I	2	
CO3	Modeling using UML structural diagrams		
M3.01	Modeling using class and object diagram	4	Applying
M3.02	Modeling using component and deployment diagram	5	Applying
CO4	Develop a prototype of the product as a team		

M4.01	Implement the application using appropriate technology (php-mysql)	8	Applying
	Series Test –II	2	

Note: Experiments shall be conducted such that all COs are attained.

Text / Reference

T/R	Book Title/Author
T1	Software Engineering, A practitioner's Approach- Roger S. Pressman, 6th edition, Mc Graw Hill International Edition.
T2	Slberschatz A., H. F. Korth and S. Sudarshan, Database System Concepts, 6/e, McGraw Hill, 2011.
T3	Elmasri R. and S. Navathe, Database Systems: Models, Languages, Design and Application Programming, Pearson Education, 2013

Online Resources

Sl.No	Website Link
1	https://staruml.io/
2	https://www.w3schools.com/sql/
3	https://sqlbolt.com/